Nmap Service Version and Operating System Detection

Aim

To execute and display Nmap commands in Kali Linux for detecting service versions, performing an aggressive scan, and identifying the operating system of a target host.

Requirements

Kali Linux, Nmap, a terminal, and a target system that you own or have explicit permission to scan. Nmap is commonly pre-installed in Kali Linux.

Important Safety Note

Run scans only against your own machine, an authorized lab VM, or a target for which you have permission. The examples below use 127.0.0.1 (the local machine). Replace it only with an authorized lab target IP.

Procedure

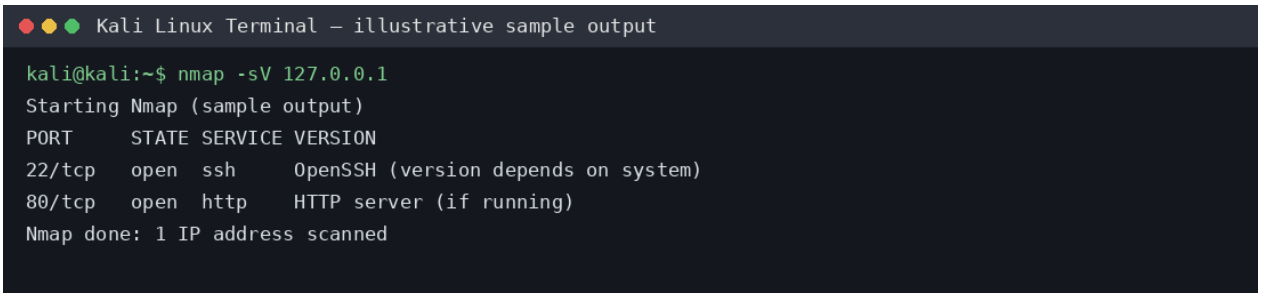
1. Open Kali Linux and launch the Terminal.
2. Check that Nmap is installed by running: nmap --version
3. Run the service-version detection command: nmap -sV 127.0.0.1
4. Run the aggressive scan command: nmap -A 127.0.0.1
5. Run operating-system detection with administrator privileges: sudo nmap -O 127.0.0.1
6. Observe the terminal output. Results depend on which services are running and the target's network configuration.

i) Detect the Version of Services Running

Command:

nmap -sV 127.0.0.1

Explanation: The -sV option asks Nmap to probe open ports and identify the service and, when possible, its version. Only services reachable on the target can be identified.



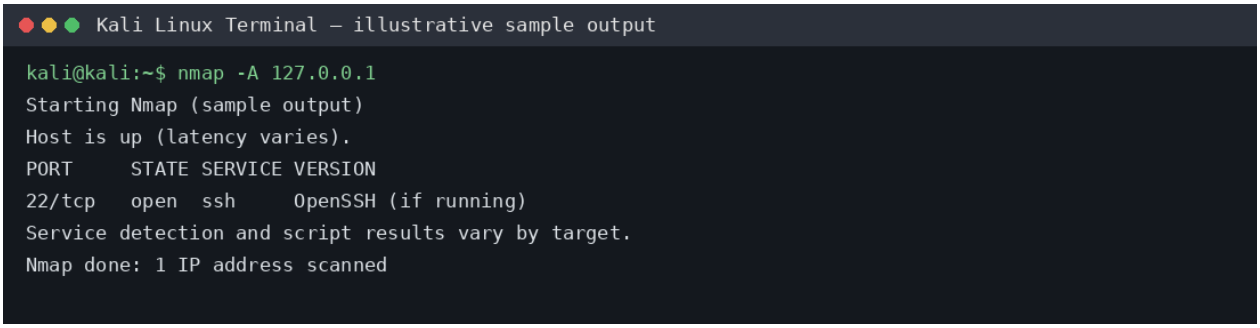
Illustrative sample output only — this is not a screenshot of a live scan. Your output may be different.

ii) Aggressive Scan

Command:

nmap -A 127.0.0.1

Explanation: The -A option enables several advanced detection features, including OS detection, service/version detection, default script scanning, and traceroute. It is more intrusive than a basic scan, so use it only on authorized targets.



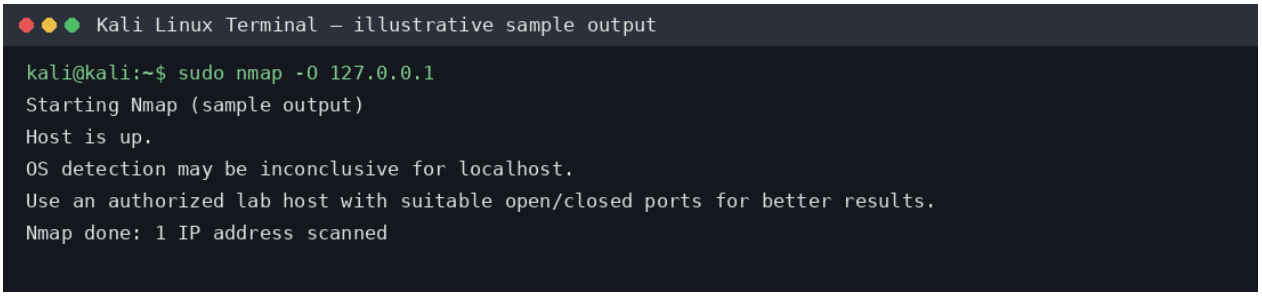
Illustrative sample output only — actual results vary by host and installed services.

iii) Detect the Operating System of the Target

Command:

sudo nmap -O 127.0.0.1

Explanation: The -O option attempts to identify the target operating system by analyzing network responses. OS detection may be inconclusive, especially when scanning localhost or when the target does not provide enough network-response information.



Illustrative sample output only — replace with your own terminal screenshot after running the command.

Result

Nmap commands for service-version detection, aggressive scanning, and operating-system detection were studied. The commands should be executed only against authorized systems, and the observed results should be recorded from the user's own Kali Linux terminal.